

M^5 Day 3: Lines

Name: _____

1. Give an equation of a line that
 - a. passes through the point $(1, 4)$. Is this line unique?
 - b. passes through the points $(1, 4)$ and $(-2, -7)$. Is this line unique?
 - c. passes through the points (a, b) and (c, d) .
 - d. has slope $m = -3$. Is this line unique?
 - e. has slope $m = -3$ and passing through the point $(-6, 5)$. Is this unique?

2. State whether each statement is True or False. If False, give an example showing why.

a. Two distinct points in the plane determine a unique line.

b. Two lines with the same slope are perpendicular.

c. Every line has an equation of the form $y = mx + b$.

d. If two lines have the same y -intercept, then they have the same slope.

e. Every horizontal line has slope 0.

3. Give an example of a line that

a. passes through exactly one of the points $(0, 0)$, $(1, 1)$, $(2, 2)$.

b. passes through all three of the points $(0, 0)$, $(1, 1)$, $(2, 2)$.

4. Find all values of k for which the points $(0, 1)$, $(2, k)$, $(4, 9)$ lie on the same line.

5. Is it possible to have a line such that passes through $(0, 0)$ and $(1, 2)$ but has slope 3?

6. Is it possible to have a line such that no points on that line have integer coordinates?